

Figure 12 — Causal Integration Ladder · Level III-A computational functional support

Per-gene synthesis of CRISPR Chronos dependency · DEMETER2 RNAi · GDSC drug-IC50 correlations · LINCS L1000 signature reversal · CellxGene Census cell-type compartment

Gene	CRISPR (Chronos cervical)	RNAi (DEMETER2 cervical)	Drug response (GDSC1+2)	LINCS L1000 reversers	scRNA-seq compartment	CIL Level III-A
CDC20	-2.74 Strong dependency	-0.26 Not dependent	9 drugs top: Cisplatin	panel rev. MG-132/olaparib/dasatinib	Progenitor / cycling migratory enteric neur	Strong Level III-A
KRT6A	-0.55 Dependency	+0.01 Not dependent	16 drugs top: Oxaliplatin	panel rev. MG-132/olaparib/dasatinib	epithelial stratified epithelial	Strong Level III-A
CDKN2A	-0.08 Not dependent	+0.16 Not dependent	9 drugs top: Palbociclib	panel rev. MG-132/olaparib/dasatinib	epithelial secretory cell	Moderate Level III-A
KRT5	-0.21 Not dependent	+0.04 Not dependent	7 drugs top: Oxaliplatin	panel rev. MG-132/olaparib/dasatinib	epithelial hair follicular kerati	Moderate Level III-A
PITX1	-0.33 Not dependent	-0.46 Not dependent	8 drugs top: GSK1059615	panel rev. MG-132/olaparib/dasatinib	epithelial squamous epithelial ce	Moderate Level III-A
S100A8	-0.19 Not dependent	-0.11 Not dependent	3 drugs top: Alpelisib	panel rev. MG-132/olaparib/dasatinib	Myeloid immature neutrophil	Moderate Level III-A
S100A9	-0.27 Not dependent	-0.08 Not dependent	17 drugs top: Palbociclib	panel rev. MG-132/olaparib/dasatinib	Myeloid immature neutrophil	Moderate Level III-A
SERPINB5	-0.15 Not dependent	-0.30 Not dependent	40 drugs top: Oxaliplatin	panel rev. MG-132/olaparib/dasatinib	epithelial granular cell of epide	Moderate Level III-A

Strong (CRISPR < −1.0 / ≥10 drug hits / on-pathway compartment) Moderate (CRISPR < −0.5 / 3–9 drug hits / supportive compartment) Limited (1–2 drug hits / off-pathway compartment) Not significant